Clinical Exome Sequencing Analysis

Patient name	**:**	**Fetus of XX**	**PIN**	**:**	**XX**
Gestational Age	**:**	**13 Weeks**	**Sample number**	**:**	**XX**
Hospital/Clinic	**:**	**XX**	**Sample collection date**	**:**	**NA**
Specimen	**:**	**Chorionic Villus**	**Sample received date**	**:**	**NA**
			Report date	**:**	**NA**

(This test confers to the PCPNDT act and does not determine the sex of the fetus)

Clinical history

XX was married non-consanguineous and has an ongoing pregnancy (LMP - 05/04/2025) conceived by natural conception. USG at 13 weeks GA showed normal findings. XX presented with history of recurrent fractures and blue sclera. XX genetic testing (Done elsewhere; 2022) showed XX harbours two heterozygous variants, a likely pathogenic variant c.2667+1G>A in intron 38 of *COL1A1* gene causative of Osteogenesis imperfecta, type I and a variant of uncertain significance c.3865G>T in exon 51 of *COL1A2* gene causative of Combined osteogenesis imperfecta and Ehlers-Danlos syndrome 2, both conditions follow autosomal dominant inheritance. She has a obstetric history of first-born male child, Master. XX, asymptomatic, alive and well, second pregnancy was terminated due to invasive prenatal genetic testing via amniocentesis (Done elsewhere; 2022) by Sanger sequencing for the variant c.2667+1G>A in *COL1A1* gene showed the presence of the variant in heterozygous state (XX fetus), third pregnancy was a missed abortion and fourth pregnancy was terminated due to invasive prenatal genetic testing (Done elsewhere; 2024) for the same variant tested in before pregnancy showed the presence of the variant in heterozygous state (XX fetus). XX mother has complaints of recurrent fractures and blue sclera (no genetic testing done). Her sister XX, also has similar complaints, genetic testing in her showed heterozygous pathogenic variant 2667+1G>A in *COL1A1* gene. Her nephew, Baby. XX presented with recurrent bone fractures, bluish discolouration of the sclera and extramedullary nails, genetic testing indicative of same heterozygous pathogenic variant found in his mother. The Fetus of XX has been evaluated for pathogenic variations.

Results

Heterozygous Variant of Uncertain Significance was identified in *COL1A2* gene

List of uncertain significant variant identified:

Gene	Region	Variant*	Allele Status	Disease	Classification*	Inheritance pattern
COL1A2 (+)	Exon 51	c.3865G>T **(p.Ala1289Ser)**	Heterozygous	Combined osteogenesis imperfecta and Ehlers-Danlos syndrome 2 **(OMIM#619120)** Ehlers-Danlos syndrome, arthrochalasia type, 2 **(OMIM#617821)** Osteogenesis imperfecta, type II **(OMIM#166210)** Osteogenesis imperfecta, type III **(OMIM#259420)** Osteogenesis imperfecta, type IV **(OMIM#166220)**	**Uncertain Significance (PM2)**	Autosomal Dominant
				Ehlers-Danlos syndrome, cardiac valvular type **(OMIM#225320)**		Autosomal Recessive

Previously reported heterozygous uncertain significant variant c.3865G>T in *COL1A2* gene in mother – XX (done elsewhere, 2022) has been detected in Fetus of XX in heterozygous state.

Reproductive decisions based on variants of uncertain significance are not recommended.

*Genetic test results are based on the recommendation of American college of Medical Genetics [1].
No other variant that warrants to be reported for the given clinical indication was identified.

Previously reported heterozygous likely pathogenic variant c.2667+1G>A in *COL1A1* gene in mother – XX (done elsewhere, 2022) has not been detected in Fetus of XX in this assay.

Interpretation

COL1A2: c.3865G>T

Variant summary: A heterozygous missense variation in exon 51 of the *COL1A2* gene (chr7:g.94429341G>T, NM_000089.4, Depth: 234x) that results in the amino acid substitution of Serine for Alanine at codon 1289 (p.Ala1289Ser) was detected.

Population frequency: This variant has a minor allele frequency of 0.00006% in gnomAD database and has not been reported in 1000 genomes database.

Clinical evidence: This variant has been classified as uncertain significance in ClinVar database [3].

***In-silico* prediction**: The *in-silico* prediction of the variant is deleterious by MutationTaster2. The reference codon is conserved across mammals in PhyloP and GERP++ tools.

OMIM phenotype: Combined osteogenesis imperfecta and Ehlers-Danlos syndrome 2 (OMIM#619120), Ehlers-Danlos syndrome, arthrochalasia type, 2 (OMIM#617821), Osteogenesis imperfecta, type II (OMIM#166210), Osteogenesis imperfecta, type III (OMIM#259420) and Osteogenesis imperfecta, type IV (OMIM#166220) are caused by heterozygous mutation in the *COL1A2* gene (OMIM*120160). Combined osteogenesis imperfecta and Ehlers-Danlos syndrome-2 (OIEDS2) is an autosomal dominant generalized connective tissue disorder characterized by features of both osteogenesis imperfecta (bone fragility, long bone fractures, blue sclerae) and Ehlers-Danlos syndrome (joint hyperextensibility, soft and hyperextensible skin, abnormal wound healing, easy bruising, vascular fragility). These diseases follow autosomal dominant pattern of inheritance [2]. Ehlers-Danlos syndrome, cardiac valvular type (OMIM#225320) is caused by homozygous or compound heterozygous mutation in the *COL1A2* gene (OMIM*120160). This disease follows autosomal recessive pattern of inheritance [2].

Variant classification: Based on the evidence, this variant is classified as variant of uncertain significance. **In this view, clinical correlation and familial segregation analysis are strongly recommended to establish the significance of the finding. If the results do not correlate, additional testing may be considered based on the phenotype observed**.

MCC Status

- The sample has been screened for maternal cell contamination and it is found to be negative for MCC (Refer Appendix: I).

Recommendations

- **Reproductive decisions based on variants of uncertain significance are not recommended.**
- **Alternative test is strongly recommended to rule out the deletion/duplication.**
- Genetic counselling is recommended.

Methodology

DNA extracted from the chorinic villus was used to perform targeted gene capture using a custom capture kit. The targeted libraries were sequenced to a targeted depth of 80 to 100X using SURFSeq 5000 sequencing platform. This kit has deep exonic coverage of all the coding regions including the difficult to cover regions. The sequences obtained are aligned to human reference genome (GRCh38.p13) using Sentieon aligner and analyzed using Sentieon for removing duplicates, recalibration and re-alignment of indels. Sentieon DNAscope has been used to call the variants. Detected variants were annotated and filtered using the VarSeq software with the workflow implementing the ACMG guidelines for variant classification. The variants were annotated using 1000 genomes (V2), gnomAD (v4.1.0,3.1.2,2.1.1), ClinVar, OMIM, dbSNP, NCBI RefSeq Genes. In-silico predictions of the variant was carried out using VS-SIFT, VS-PolyPhen2, PhyloP, GERP++, GeneSplicer, MaxEntScan, NNSplice, PWM Splice Predictor. Only non-synonymous and splice site variants found in the coding regions were used for clinical interpretation. Silent variations that do not result in any change in amino acid in the coding region are not reported.

Sequence data attributes

Total reads generated	:	18.70 Gb
Data ≥ Q30	:	96.98%

Genetic test results are reported based on the recommendations of American College of Medical Genetics [1], as described below:

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Classification	Interpretation
Pathogenic	A disease-causing variation in a gene which can explain the patients' symptoms has been detected. This usually means that a suspected disorder for which testing had been requested has been confirmed
Likely Pathogenic	A variant which is very likely to contribute to the development of disease however, the scientific evidence is currently insufficient to prove this conclusively. Additional evidence is expected to confirm this assertion of pathogenicity.
Variant of Uncertain Significance	A variant has been detected, but it is difficult to classify it as either pathogenic (disease causing) or benign (non- disease causing) based on current available scientific evidence. Further testing of the patient or family members as recommended by your clinician may be needed. It is probable that their significance can be assessed only with time, subject to availability of scientific evidence.

Disclaimer

- The classification of variants of unknown significance can change over time, Anderson Diagnostics and Labs cannot be held responsible for it.
- Intronic variants, UTR, Promoter region and CNV variants are not assessed using this assay.
- Certain genes may not be covered completely, and few mutations could be missed. Variants not detected by this assay may impact the phenotype.
- The variations have not been validated by Sanger sequencing.
- The above findings and result interpretation was done based on the clinical indication provided at the time of reporting.
- It is also possible that a pathogenic variant is present in a gene that was not selected for analysis and/or interpretation in cases where insufficient phenotypic information is available.
- Genes with pseudogenes, paralog genes and genes with low complexity may have decreased sensitivity and specificity of variant detection and interpretation due to inability of the data and analysis tools to unambiguously determine the origin of the sequence data in such regions.
- Incidental or secondary findings that meet the ACMG guidelines can be given upon request [4].
- This test does not determine the sex of the fetus in adherence to the PCPNDT act.

References

1. Richards, S, et al. Standards and Guidelines for the Interpretation of Sequence Variants: A Joint Consensus Recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology. Genetics in medicine: official journal of the American College of Medical Genetics. 17.5 (2015): 405-424.
2. Amberger J, Bocchini CA, Scott AF, Hamosh A. McKusick's Online Mendelian Inheritance in Man (OMIM). Nucleic Acids Res. 2009 Jan;37(Database issue):D793-6. doi: 10.1093/nar/gkn665. Epub 2008 Oct 8.
3. https://www.ncbi.nlm.nih.gov/clinvar/variation/VCV001735635.2
4. Miller, David T., et al. "ACMG SF v3. 1 list for reporting of secondary findings in clinical exome and genome sequencing: A policy statement of the American College of Medical Genetics and Genomics (ACMG)." Genetics in Medicine 24.7 (2022): 1407-1414.

This report has been reviewed and approved by:

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Appendix I

Estimation of maternal cell contamination %

No	Marker	Maternal genotype	Fetal genotype	No. of shared alleles	MCC	Remarks
1	D3S1358	15, 17	15	1	-	Informative
2	TH01	8	8, 9	1	-	Non informative
3	D21S11	32.2, 33.2	30, 33.2	1	-	Informative
4	D18S51	13, 16	13, 18	1	-	Informative
5	Penta E	9, 13	9, 16	1	-	Informative
6	D5S818	11	10, 11	1	-	Non informative
7	D13S317	12	8, 12	1	-	Non informative
8	D7S820	8, 10	10, 12	1	-	Informative
9	D16S539	11	11, 13	1	-	Non informative
10	CSF1PO	10, 12	10	1	-	Informative
11	Penta D	9, 11	9, 10	1	-	Informative
12	Amelogenin	X	Not revealed	-	-	-
13	vWA	15, 18	15, 16	1	-	Informative
14	D8S1179	10, 13	10, 13	2	-	Non informative
15	TPOX	8, 11	11	1	-	Informative
16	FGA	22, 23	22, 26	1	-	Informative
The average percentage of maternal cell contamination in the fetal sample is 0%						